

### Objectifs :

Le but de ce TP est de :

- ✓ Concevoir le plan de partitionnement LVM des disques d'un système Linux.
- ✓ Formater des partitions et Gérer les systèmes de fichiers.
- ✓ Contrôler le montage et le démontage d'un système de fichiers.
- ✓ Gérer les quotas disque.

### Environnement de travail :

- ✓ Machine virtuelle ou physique avec un système d'exploitation Ubuntu 22.04

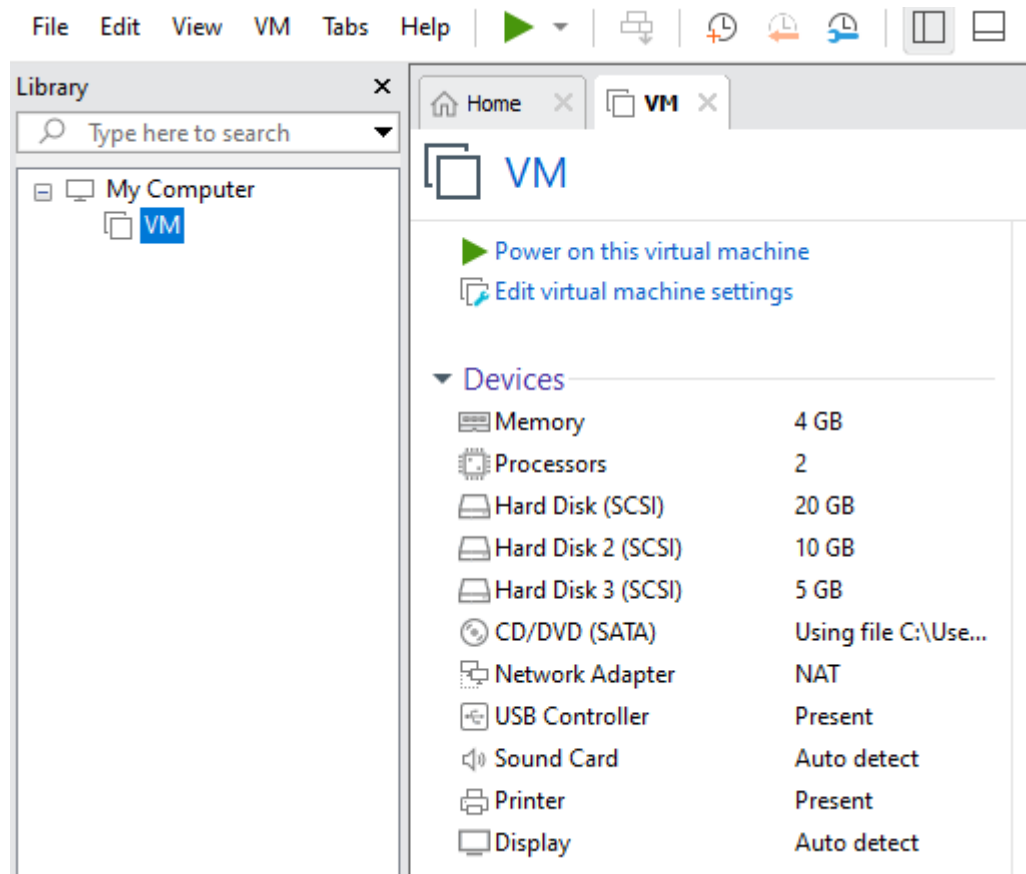
### Travail demandé :

#### Partie I : Gestion des disques

1. Ajouter deux disques durs virtuels de tailles respectivement 10Go et 5Go

Depuis votre machine virtuelle :

- Aller à « Settings », puis cliquer sur « Add »
- Choisir « Hard Disk », appuyer sur Next..... et spécifier la taille du disque à ajouter
- Après ajout des deux disques, on obtient un affichage comme indiqué ci-dessous



2. Vérifier la création de ces deux disques. Sous quels noms le système reconnaît les deux disques ?

```
esprit@esprit-virtual-machine:~$ sudo fdisk -l
[sudo] password for esprit:

Disk /dev/sda: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: E2D880AB-2F25-4BA9-BC74-4FA7CBFE755A

Device            Start      End  Sectors  Size Type
/dev/sda1          2048      4095    2048    1M BIOS boot
/dev/sda2          4096 1054719 1050624  513M EFI System
/dev/sda3        1054720 41940991 40886272 19.5G Linux filesystem

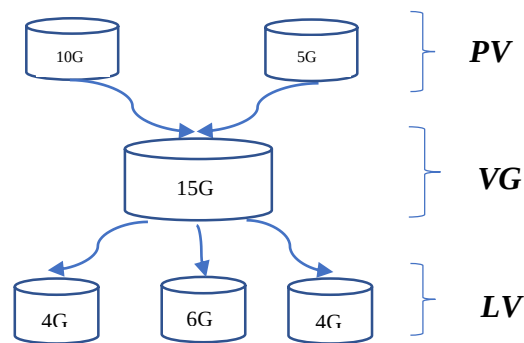
Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdc: 5 GiB, 5368709120 bytes, 10485760 sectors
Disk model: VMware Virtual S
```

3. Reproduire le schéma de partitionnement LVM suivant :

Le package LVM n'est par défaut installé, on doit tout d'abord l'installer

```
esprit@esprit-virtual-machine:~$ sudo apt-get install lvm2
```



```
esprit@esprit-virtual-machine:~$ sudo pvcreate /dev/sdb /dev/sdc
Physical volume "/dev/sdb" successfully created.
Physical volume "/dev/sdc" successfully created.
```

```
esprit@esprit-virtual-machine:~$ sudo vgcreate vg1 /dev/sdb /dev/sdc
Volume group "vg1" successfully created
```

```
esprit@esprit-virtual-machine:~$ sudo vgdisplay vg1
--- Volume group ---
VG Name                vg1
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   1
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                0
Open LV               0
Max PV                 0
Cur PV                2
Act PV                2
VG Size                14.99 GiB
PE Size                4.00 MiB
Total PE              3838
Alloc PE / Size        0 / 0
Free PE / Size         3838 / 14.99 GiB
VG UUID                gN6YsC-kQRH-l8ON-l0RC-WGUC-CVf1-CySrM7
```

```
esprit@esprit-virtual-machine:~$ sudo lvcreate -n vol1 -L 4g vg1
Logical volume "vol1" created.
esprit@esprit-virtual-machine:~$ sudo lvcreate -n vol2 -L 6g vg1
Logical volume "vol2" created.
esprit@esprit-virtual-machine:~$ sudo lvcreate -n vol3 -L 4g vg1
Logical volume "vol3" created.
```

```

esprit@esprit-virtual-machine:~$ sudo lvdisplay vg1
--- Logical volume ---
LV Path                /dev/vg1/vol1
LV Name                 vol1
VG Name                 vg1
LV UUID                 DCwufH-Q640-5ReJ-gCDe-r30t-yUdI-UiP3j0
LV Write Access         read/write
LV Creation host, time esprit-virtual-machine, 2022-11-15 04:23:00 +0100
LV Status                available
# open                  0
LV Size                 4.00 GiB
Current LE              1024
Segments                1
Allocation               inherit
Read ahead sectors      auto
- currently set to     256
Block device            253:0

--- Logical volume ---
LV Path                /dev/vg1/vol2
LV Name                 vol2
VG Name                 vg1
LV UUID                 Tb05fd-KA9S-m6Xg-qtcG-WmTf-mlon-EnKAeo
LV Write Access         read/write
LV Creation host, time esprit-virtual-machine, 2022-11-15 04:23:14 +0100
LV Status                available
# open                  0
LV Size                 6.00 GiB
Current LE              1536
Segments                2
Allocation               inherit
Read ahead sectors      auto
- currently set to     256
Block device            253:1

--- Logical volume ---
LV Path                /dev/vg1/vol3
LV Name                 vol3
VG Name                 vg1
LV UUID                 FaYrkf-B0bY-JPBZ-yJIW-cE1l-wtLt-pgLjIW
LV Write Access         read/write
LV Creation host, time esprit-virtual-machine, 2022-11-15 04:23:28 +0100
LV Status                available
# open                  0
LV Size                 4.00 GiB
Current LE              1024
Segments                1
Allocation               inherit
Read ahead sectors      auto
- currently set to     256
Block device            253:2

```

4. Formater les volumes logiques avec les systèmes de fichiers comme suit :
  - a. 1<sup>er</sup> volume logique : ext2

```
esprit@esprit-virtual-machine:~$ sudo mkfs -t ext2 /dev/vg1/vol1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1048576 4k blocks and 262144 inodes
Filesystem UUID: 59889f90-b133-451e-a138-03843f9028e1
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Writing superblocks and filesystem accounting information: done
```

b. 2<sup>ème</sup> volume logique : ext4

```
esprit@esprit-virtual-machine:~$ sudo mkfs -t ext4 /dev/vg1/vol2
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1572864 4k blocks and 393216 inodes
Filesystem UUID: d136a718-ef5a-411f-81b5-78ee9799c151
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

c. 3<sup>ème</sup> volume logique : ext3

```
esprit@esprit-virtual-machine:~$ sudo mkfs -t ext3 /dev/vg1/vol3
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1048576 4k blocks and 262144 inodes
Filesystem UUID: 095e3c2c-6dca-49ef-a8a1-a058e53394d1
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

5. Sur le 1<sup>er</sup> volume logique, remplacer ext2 par ext3

```
esprit@esprit-virtual-machine:~$ sudo tune2fs -j /dev/vg1/vol1
tune2fs 1.46.5 (30-Dec-2021)
Creating journal inode: done
```

6. Configurer le système de telle sorte que :

- a. Les fichiers de journalisation seront montés sur le premier volume logique

```

esprit@esprit-virtual-machine:~$ sudo mount /dev/vg1/vol1 /var/log
esprit@esprit-virtual-machine:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           390M  1.9M  388M   1% /run
/dev/sda3       20G   9.3G   8.8G  52% /
tmpfs           2.0G    0  2.0G   0% /dev/shm
tmpfs           5.0M  4.0K   5.0M   1% /run/lock
/dev/sda2       512M   5.3M  507M   2% /boot/efi
tmpfs           390M  2.4M  387M   1% /run/user/1000
/dev/sr0        3.6G  3.6G    0 100% /media/esprit/Ubuntu 22.04.1 LTS am
d64
/dev/mapper/vg1-vol1 4.0G   24K   3.8G   1% /var/log

```

- b. Les répertoires personnels seront montés sur le second volume logique

```

esprit@esprit-virtual-machine:~$ sudo mount /dev/vg1/vol2 /home
esprit@esprit-virtual-machine:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           390M  1.9M  388M   1% /run
/dev/sda3       20G   9.3G   8.8G  52% /
tmpfs           2.0G    0  2.0G   0% /dev/shm
tmpfs           5.0M  4.0K   5.0M   1% /run/lock
/dev/sda2       512M   5.3M  507M   2% /boot/efi
tmpfs           390M  2.4M  387M   1% /run/user/1000
/dev/sr0        3.6G  3.6G    0 100% /media/esprit/Ubuntu 22.04.1 LTS am
d64
/dev/mapper/vg1-vol1 4.0G   24K   3.8G   1% /var/log
/dev/mapper/vg1-vol2 5.9G   24K   5.6G   1% /home

```

- c. Les fichiers temporaires seront montés sur le troisième volume logique tout en empêchant l'exécution des binaires.

```

esprit@esprit-virtual-machine:~$ sudo mount /dev/vg1/vol3 /tmp
esprit@esprit-virtual-machine:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           390M  1.9M  388M   1% /run
/dev/sda3       20G   9.3G   8.8G  52% /
tmpfs           2.0G    0  2.0G   0% /dev/shm
tmpfs           5.0M  4.0K   5.0M   1% /run/lock
/dev/sda2       512M   5.3M  507M   2% /boot/efi
tmpfs           390M  2.4M  387M   1% /run/user/1000
/dev/sr0        3.6G  3.6G    0 100% /media/esprit/Ubuntu 22.04.1 LTS am
d64
/dev/mapper/vg1-vol1 4.0G   24K   3.8G   1% /var/log
/dev/mapper/vg1-vol2 5.9G   24K   5.6G   1% /home
/dev/mapper/vg1-vol3 3.9G   92K   3.7G   1% /tmp

```

7. Faire en sorte que les nouveaux volumes logiques seront montés automatiquement au démarrage du système



```
GNU nano 6.2 /etc/fstab *
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda3 during installation
UUID=9881117b-dbec-4cb6-a49c-f883b395a827 / ext4 errors=remou>
# /boot/efi was on /dev/sda2 during installation
UUID=2AC8-0A63 /boot/efi vfat umask=0077 0 1
/swapfile none swap sw >
/dev/vg1/vol1 /var/log ext3 defaults 0 0
/dev/vg1/vol2 /home ext4 defaults 0 0
/dev/vg1/vol3 /tmp ext3 noexec 0 0

esprit@esprit-virtual-machine:~$ sudo mount -av
/ : ignored
/boot/efi : already mounted
none : ignored
/var/log : successfully mounted
/home : successfully mounted
/tmp : successfully mounted
```

## Partie II : Gestion des quotas

1. Créer deux utilisateurs « user1 » et « user2 » appartenant à un groupe « Users »

```
esprit@esprit-virtual-machine:~$ sudo groupadd Users
esprit@esprit-virtual-machine:~$ sudo useradd -g Users user1
esprit@esprit-virtual-machine:~$ sudo useradd -g Users user2
```

2. Créer deux utilisateurs « admin1 » et « admin2 » appartenant à un groupe « Admins »

```
esprit@esprit-virtual-machine:~$ sudo groupadd Admins
esprit@esprit-virtual-machine:~$ sudo useradd -g Admins admin1
esprit@esprit-virtual-machine:~$ sudo useradd -g Admins admin2
```

3. Faire en sorte que « user1 » ne pourra créer que **20 fichiers** sur son répertoire personnel. En cas de dépassement de limite, il bénéficiera encore de **3 jours** pour pouvoir créer **5 autres fichiers**.

Installer le package « quota » s'il n'existe pas

```
esprit@esprit-virtual-machine:~$ sudo apt-get install quota
```

```

GNU nano 6.2 /etc/fstab *
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda3 during installation
UUID=8ddb544a-84af-4426-8687-33b74c67abf1 / ext4 errors=remount
# /boot/efi was on /dev/sda2 during installation
UUID=C53F-388B /boot/efi vfat umask=0077 0 1
/swapfile none swap sw 0 0
/dev/vg1/vol1 /var/log ext3 defaults 0 0
/dev/vg1/vol2 /home ext4 defaults,usrquota,grpquota 0
/dev/vg1/vol3 /tmp ext3 noexec 0 0

```

```
esprit@esprit-virtual-machine:/home$ sudo mount -o remount /home
```

```

esprit@esprit-virtual-machine:/home$ sudo quotacheck -aucvgm
quotacheck: Your kernel probably supports ext4 quota feature but you are using e
xternal quota files. Please switch your filesystem to use ext4 quota feature as
external quota files on ext4 are deprecated.
quotacheck: Scanning /dev/mapper/vg1-vol2 [/home] done
quotacheck: Cannot stat old user quota file /home/aquota.user: No such file or d
irectory. Usage will not be subtracted.
quotacheck: Cannot stat old group quota file /home/aquota.group: No such file or
directory. Usage will not be subtracted.
quotacheck: Cannot stat old user quota file /home/aquota.user: No such file or d
irectory. Usage will not be subtracted.
quotacheck: Cannot stat old group quota file /home/aquota.group: No such file or
directory. Usage will not be subtracted.
quotacheck: Checked 3 directories and 0 files
quotacheck: Old file not found.

```

```

esprit@esprit-virtual-machine:/home$ sudo quotaon -av
quotaon: Your kernel probably supports ext4 quota feature but you
ernal quota files. Please switch your filesystem to use ext4 quota
ernal quota files on ext4 are deprecated.
/dev/mapper/vg1-vol2 [/home]: group quotas turned on
/dev/mapper/vg1-vol2 [/home]: user quotas turned on

```

```
esprit@esprit-virtual-machine:/home$ sudo edquota -u user1
```

```

File Edit View Search Terminal Help
GNU nano 6.2 /tmp//EdP.arRZKi4 *
Disk quotas for user user1 (uid 1001):
Filesystem blocks soft hard inodes soft hard
/dev/mapper/vg1-vol2 0 0 0 0 20 0

```

```

GNU nano 6.2 /tmp//EdP.aME4vHW *
Times to enforce softlimit for user user1 (uid 1001):
Time units may be: days, hours, minutes, or seconds
Filesystem block grace inode grace
/dev/mapper/vg1-vol2 unset 3days

```

4. Faire en sorte que les membres du groupe « Admins » pourront partager uniquement jusqu'à **2Go d'espace disque** et auront un message de notification lorsqu'ils dépassent **1.5Go**.

```
esprit@esprit-virtual-machine:/home$ sudo edquota -g Admins
```



```
File Edit View Search Terminal Help
GNU nano 6.2 /tmp//EdP.amDi8kl *
Disk quotas for group Admins (gid 1002):
Filesystem blocks soft hard inodes soft hard
/dev/mapper/vg1-vol2 0 1572864 2097152 0 0 0

esprit@esprit-virtual-machine:/home$ su admin1
Password:
$ dd if=/dev/zero of=testfile bs=100M count=16
16+0 records in
16+0 records out
1677721600 bytes (1.7 GB, 1.6 GiB) copied, 14.5087 s, 116 MB/s
$ dd if=/dev/zero of=testfile bs=100M count=17
17+0 records in
17+0 records out
1782579200 bytes (1.8 GB, 1.7 GiB) copied, 1.80282 s, 989 MB/s
$ dd if=/dev/zero of=testfile bs=100M count=18
18+0 records in
18+0 records out
1887436800 bytes (1.9 GB, 1.8 GiB) copied, 1.79061 s, 1.1 GB/s
$ dd if=/dev/zero of=testfile bs=100M count=20
20+0 records in
20+0 records out
2097152000 bytes (2.1 GB, 2.0 GiB) copied, 2.14795 s, 976 MB/s
$ dd if=/dev/zero of=testfile bs=100M count=21
dd: error writing 'testfile': Disk quota exceeded
21+0 records in
20+0 records out
2147479552 bytes (2.1 GB, 2.0 GiB) copied, 2.20228 s, 975 MB/s
```

5. Afficher un rapport détaillé sur les quotas alloués aux différents utilisateurs et groupes.

```
esprit@esprit-virtual-machine:/home$ sudo repquota -ug /home
*** Report for user quotas on device /dev/mapper/vg1-vol2
Block grace time: 7days; Inode grace time: 7days
      Block limits
User      used  soft  hard  grace  used  soft  hard  grace
-----
root      --    20    0    0      2    0    0
esprit    --  12744  0    0      43    0    0
user1     +-    0    0    0      26   20    0  6days
admin1    -- 2097152  0    0      1    0    0

*** Report for group quotas on device /dev/mapper/vg1-vol2
Block grace time: 7days; Inode grace time: 7days
      Block limits
Group     used  soft  hard  grace  used  soft  hard  grace
-----
root      --    20    0    0      2    0    0
esprit    --  12744  0    0      43    0    0
Users     --    0    0    0      26    0    0
Admins    +- 2097152 1572864 2097152 6days  1    0    0
```